

Tutorial Sheet 1

Solutions

Question (1)

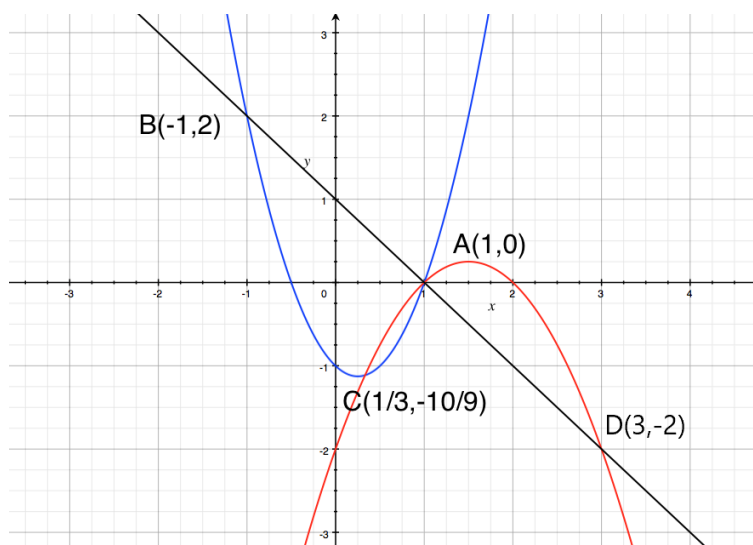
- (a) $x_1 = \frac{1}{2}, x_2 = 1$ (b) No solutions in $\mathbb{R}$
(c) $x_1 = x_2 = 2$ (Repeated solution) (d) $x_1 = -\frac{2}{3}, x_2 = 1$

Question (2)

- (a) $x_1 = 1$ is the only solution in $\mathbb{R}$ (b) $x_1 = -3, x_2 = -2, x_3 = 2$
(c) $x_1 = -2, x_2 = 1, x_3 = 3$ (d) $x_1 = 2, x_2 = -1 - \sqrt{5}, x_3 = -1 + \sqrt{5}$

Question (3)

- (a) $x \leq -9$ (b) $x \in \left(-\infty, -\frac{5}{2}\right) \cup (-1, \infty)$ (c) $x \in (-\infty, -3) \cup (-2, \infty)$

Question (4)

Intersection points are: $A(1, 0)$, $B(-1, 2)$, $C(1/3, -10/9)$ and $D(3, -2)$. Parabola $y = 2x^2 - x - 1$ is in blue, parabola $y = -x^2 + 3x - 2$ is in red, line $y = -x + 1$ is in black.

Note: to find the x -coordinates of points A , C , solve the equation

$$2x^2 - x - 1 = -x^2 + 3x - 2$$

and substitute the values of x found into the equation of either of the two parabolas to get the corresponding y -coordinates. To find the x -coordinate of B , solve the equation

$$2x^2 - x - 1 = -x + 1$$

and substitute one of the two solutions, i.e. $x = -1$, into the equation of the line. The other solution for the equation above is $x = 1$, which will give you point A again.

To find point D , repeat the latter process but use the other quadratic expression on the LHS instead.

Question (5)

$$(a) \quad \left(x + \frac{1}{2}\right)^2 - \frac{1}{4} \quad (b) \quad \left(x - \frac{3}{2}\right)^2 - \frac{5}{4} \quad (c) \quad x^2 + 1$$

Note: question (c) is already in the “standard” quadratic form, i.e.

$$x^2 + 1 = (x + 0)^2 + 1.$$

Other “unusual” solutions to question (c) are possible, for example $(x + 1)^2 - 2x$. These types of solutions are unusual because they introduce a term proportional to x . This is non-standard. We can always rewrite any quadratic in infinitely many different ways if we introduce in the new expression a term linear in x . For instance, $(x + 1)^2 - 2x$, $(x - 1)^2 - 2x$, $(x + 2)^2 - 4x - 3$, ... are all equivalent different ways of writing $x^2 + 1$.

Question (6)

$$(a) \ (x+7) + \frac{4}{x-1} \quad (b) \ (2x-12) + \frac{77x-35}{x^2+6x-3} \quad (c) \ (x^2-3x+5) + \frac{-18x+6}{x^2+3x-1}$$

Note: Question (b) requires a bit of patience with the calculations.

Question (7)

$$\begin{array}{ll} (a) \ x = e^4 & (b) \ x = 2 \\ (c) \ x_1 = 0, \ x_2 = -1 & (d) \ x = \sqrt[3]{\ln 10} = (\ln(10))^{\frac{1}{3}} \\ (e) \ x = \frac{1}{4}(\sqrt{17} - 5) & (f) \ x_1 = 1, \ x_2 = 4 \end{array}$$

Note: In question (e), one of the solutions of the quadratic equation is not acceptable, because the argument of any logarithm must always be positive.

Question (8)

$$\exp(x \ln x), \quad \exp(\cos x \ln(\sin x)), \quad \exp(2 \tan x \ln x).$$

Question (9)

$$\frac{\log_2 7}{\log_2 3}, \quad \frac{\log_2 2x}{\log_2 10}, \quad \frac{\log_2 3}{\log_2(x^2+1)}.$$

Note: The answer to question (b) can be further simplified to $\frac{1 + \log_2 x}{\log_2 10}$. Both answers are acceptable and would attract full marks.